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Title: Comparative analysis of transmission and vaccination efficacy in Omicron and Delta variant outbreaks in China

Abstract

Currently circulating variants of concern of SARS-Cov-2 include the Delta and Omicron variants, which include BA.1, BA.2, and descendent lineages, posing a new challenge for transmission. Delta and Omicron BA.1 have been replaced by Omicron BA.2 worldwide. Comparing Delta, Omicron BA.1, and Omicron BA.2 is important, as SARS-Cov-2 has been shown to be mutating faster. We selected three outbreaks caused by difference variants in China and estimated epidemiological parameters and vaccine effectiveness. Omicron BA.2 was transmitted faster than BA.1 and Delta, with a shorter serial interval (2.24 days *vs.* 3.00 days *vs.* 5.70 days) and larger effective reproductive number (3.56 *vs.* 2.94 *vs.* 1.93), although full vaccine and booster dose coverage was increased. Vaccine effectiveness against Omicron BA.1 and BA.2 infection was not significantly different in regression modeling. In the event of a future pandemic, enhancing booster vaccination rate and increasing treatment availability are needed to avert preventable mortality.

Introduction

Variants of Concern (VOCs) have raise new transmission, severity, and vaccination response concerns during the Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) pandemic[1]. Omicron variant (B.1.1.529) of SARS-CoV-2, which was

first identified in South Africa on 24 November 2021[2], has been replacing the Delta variant rapidly worldwide[3, 4] and causing many breakthrough infections. It has been previously shown that the Omicron variant may be more transmissible than the Delta variant[5, 6, 7], with a growth rate 3.3 times faster and an effective reproduction number (R_{eff}) 4.2 times higher. According to some studies, Omicron infection conveys a decreased hospitalization and mortality risk as compared to the Delta variant[8, 9]. COVID-19 vaccination is less effective against Omicron than it is against Delta, and the protection it conveys against Omicron wears off faster[10]. Infectivity, vaccination effectiveness, and mortality risk were all significantly different between the Omicron and Delta variants. A comparison of Omicron and Delta variations is still being investigated, particularly in terms of transmission, symptomatology, and vaccination effects in the real world.

Despite China's high vaccination rate (86.73 percent fully vaccinated and 52.82 percent vaccinated with boosters as of May 12, 2022)[11], poor vaccination efficacy and immunity escape continue to produce a significant number of COVID-19 infections in cities such as Hong Kong and Shanghai[12, 13]. China's vaccination effort and its efficacy entails more investigation to prevent, or at least mitigate, future outbreaks. China's "dynamic zero COVID" policy was instrumental in quickly halting the transmission of the original virus, as well as its Delta and Omicron variants, after the Wuhan epidemic[14]. The transmission characteristics of infection in Delta and Omicron mutant strains were not fully defined. The goal of this study is to examine the characteristics of Delta and Omicron variants, including transmission, Ct value, and

vaccine efficacy, to provide additional information about the COVID-19 pandemic in China.

Results

Outbreak characteristics

We examined three outbreaks caused by varied variants of SARS-Cov-2 and located in Southern China (Fig. 1a). A total of 202 infections due to variants of SARS-Cov-2 in three outbreaks were included in our analysis: of these infections, 33 (16.34%) were caused by the Omicron BA.2 variant, 38 (18.81%) were caused by the Omicron BA.1 variant, and 129 (63.86%) were caused by the Delta variant. The Omicron BA.2 outbreak (from March 10 to April 6, 2022) was caused by an infected traveler from Shanghai city, and did not grow to become a large-scale community infection (Fig. 1b, Fig. 2a, Fig. 1e). For the Omicron BA.1 outbreak (from January 8 to January 21, 2022), the index case and imported cases from other cities were visiting the same location at the same time (Fig. 1c). A total of 16 (42.11%) infections were attributed to a super-spreader, who had accepted the third dose of CoronaVac (Sinovac Biotech Ltd.) a month prior (Fig. 1f, Fig. 2b). The Delta outbreak time period extended from July 22 until August 15, 2021, and was caused by an exposed traveler from Nanjing airport (Fig. 1d). Superspreading events were also observed in the Delta outbreak and produced a severe case of infection in a 60-year-old, unvaccinated patient (Fig. 1g, Fig. 2c). The demographic and clinical characteristics of the examined infections are provided in Table 1. The median age of infections of Omicron BA.2, Omicron BA.1

and Delta variant was varied (21.5 years *vs.* 31.0 years *vs.* 34.0 years), and 137 (67.82%) infections were in adults (19-64 years) (Fig. 3a, Fig. 3c, Fig. 3e). In addition, the proportion of asymptomatic infections were decreased from 15% in the case of the Delta outbreak to 6%-8% in those of the Omicron variants.

A total of 21716 contacts were introduced by 202 infections and tracked by local CDC health workers: an average of 109.43 contacts per infection were detected for the Delta variant, 102.53 for the Omicron BA.1 variant, and 105.80 for the Omicron BA.2 variant. The characteristics of the infections under study are shown in Table 2. Full vaccine coverage of contacts in Delta, Omicron BA.1, and BA.2 outbreaks were 44.33%, 79.77%, and 68.72%, respectively. Booster dose coverage was 40.84% in the Omicron BA.1 outbreak and 29.62% in that of Omicron BA.2. Vaccine coverage of children (0-18 years old) and older adults (65 years of age or older) were lower than that of younger adults (Fig. 3b, Fig. 3d, Fig. 3f). More than 99% of the vaccines accepted by contacts with infected individuals were produced by six manufacturers, including Sinovac Biotech Ltd., Sinopharm Group Co. Ltd., CanSino Biologics Inc., Anhui Zhifei Longcom Biopharmaceutical Co. Ltd., and Shenzhen Kangtai Biological Products Co., Ltd. The mixed vaccine strategy was also observed in each outbreak, but 98.27% of vaccinations were a dosing combination of CoronaVac (Sinovac Biotech Ltd.) and COVILO (Sinopharm Group Co. Ltd.) (Fig. 3g-i).

Epidemiological parameters of variants

The distribution of epidemiological parameters was fitted to Gamma distribution; the Lognormal distribution and Weibull distribution were used as well, and showed

similar goodness-of-fit as measured by log-likelihood (Supplementary Fig. 1). The parameters of SARS-Cov-2 variants are shown in Table 3. Compared with the Delta variant, Omicron BA.2 and BA.1 variants were transmitted with a shorter SI (2.24 days *vs.* 3.00 days *vs.* 5.70 days) (Fig. 4a), and IP (3.07 days *vs.* 4.35 days *vs.* 7.63 days) (Fig. 4b) in the outbreaks examined. The difference between SI and IP of each infected individual were showed in Fig. 4c. A percentage of 35.09% of cases where SIs were shorter than IPs was recorded for individuals infected with Omicron BA.1; similarly, 70.98% of cases were recorded as such in individuals infected with Omicron BA.2, and 75.00% of cases in those infected by Delta. The estimation of GT was based on a review of the exposed period and probable time of infection. Omicron BA.1 displayed a shorter GT than the Delta variant (1.76 days *vs.* 2.52 days), and a similar GT to Omicron BA.2 (2.93 days *vs.* 2.52 days). TG of variants, defined as the time period between positive tests for infecting and infected individuals, which did not relay on the recall of the infected individual, were also varied. Delta and Omicron BA.1 showed similar TG values (3.61 days *vs.* 3.56 days), and Omicron BA.2 displayed a TG of 1.95 days (Fig. 4e). R_{eff} of Delta at increased stage was 1.93 (95%CI: 1.3-2.72), 2.94 (95%CI: 1.41-5.30) for Omicron BA.1, and 3.56 (95%CI: 1.20-7.94) for Omicron BA.2 (Fig. 4f). We also compared IP, SI, TG, and GT values between adults (older than 19 years) and children (0-18 years), and found that children had shorter IP in the Omicron BA.1 outbreak (4.00 days *vs.* 4.89 days) and shorter TG values in both the Omicron BA.1 (3.11 days *vs.* 4.14 days) and Delta (2.37 days *vs.* 4.06 days) outbreaks (Supplementary Fig. 2). This may related to a deficient

vaccine coverage in children, in part (Fig. 3d, Fig. 3f). But the difference between the epidemiological parameters of Omicron BA.2, as opposed to other variants, cannot be explained entirely by deficient vaccine coverage in children.

Ct value and vaccine efficacy

Based on samples collected from the 202 infections, statistically significant differences between Ct values of individuals with various vaccination statuses were not observed, regardless of the presence of the N and ORF genes (Fig. 5a, Fig. 5c). There was no statistically significant difference ($P > 0.05$) between the Ct values of Delta, Omicron BA.1, and Omicron BA.2 (Fig. 5b, Fig. 5d). The interaction effects of variants and vaccine status did not produce differences between Ct values (Fig. 5e-f).

The overall vaccine efficacy against the Delta variant in fully vaccinated individuals was 51.68% (95%CI: 28.90-67.16%) (Fig. 5g). In conditional logistic regression model, the vaccine efficacy against Delta variant infection adjusted by age group was 67.87% (95%CI: 51.67-78.64%) (Fig. 5h). For Omicron BA.1 and BA.2, difference in vaccine effectiveness against infection was not observed, regardless of adjustment by age group.

Discussion

Because of the high communicability of the Omicron variant, its transmission has mostly superseded that of Delta in most nations. On December 13, 2021, the first case of Omicron infection was recorded in Tianjin City of China's mainland. The dynamic zero-COVID policy of Omicron variant had constantly dominated Chinese strategy of

prevention and infection control. Our findings imply that Omicron's transmissibility is 1.5-1.8 higher than that of Delta in terms of viral transmission. This is lower than the value reported by other studies, which claim that Omicron has a transmissibility 2.5 to 4 higher than that of Delta[7, 25]. This might be attributable to the rising rate of fully vaccinated and booster-vaccinated people. Meanwhile, the geographic variability is also linked to inconsistencies in the use of COVID-19 preventative and control measures in different regions. We also saw that the transmissibility of the two Omicron sub-lineages differed, with Omicron BA.2 being 1.2 times more transmissible than BA.1, which is similar to the results of several studies that suggest that BA.2 is 30 to 40 percent more infectious than BA.1[26, 27, 28]. In comparison to Delta, applying a dynamic zero-COVID policy for interrupting Omicron transmission may necessitate greater preventative and control efforts.

The shorter incubation period and infectious production time of Omicron were the main outcomes of our investigation of its high communicability. We found Delta to have a median incubation duration (7.63 days) longer than that of Omicron BA.1 (4.35 days) and BA.2 (3.07 days), which was consistent with the comparative results of several studies[29, 30, 31]. A study in Belgium reported that a mean serial interval of 2.75 days was detected for the Omicron variant and 3.00 days for the Delta variant[32], while the serial interval of the Delta variant (5.70 days) we observed was much higher than that of the Omicron variant (3.00 days and 2.24 days). Meanwhile, the transmissibility of the variations mentioned prior appeared before symptoms did. We found that the Delta variant became transmissible 3 days before symptom onset, but

that BA.2 became transmissible 1 day before symptoms. According to one study, individuals infected with the Delta variant may spread the virus for approximately two days before symptom onset (BSO)[33, 34], whereas Omicron transmission usually occurs 1-3 days BSO[5]. Our findings imply that controlling the outbreak is expected to be difficult if it relies just on symptom surveillance, and that close contact tracing may be the more effective option.

According to one study[35], the peak Ct value for Omicron infections was lower than for Delta infections (23.3 *vs.* 20.5). Although the Ct value of BA.2 was lower than that of Delta in our investigation, it was not significantly different. Regardless of accounting for vaccination status, there was no significant difference in Ct values depending on lineage, which was consistent with another study's comparative variation[36]. A study indicated that the Ct values in health care workers sampled within 5 days after symptom onset were significantly higher for Omicron cases than Delta cases (23.8 *vs.* 21.7, $P = 0.008$)[37]. Our study was limited to a single Ct value obtained after a case was discovered, and infections of Omicron BA.2 outbreaks were reported to progress faster than outbreaks caused by Omicron BA.1 and Delta (Supplementary Fig. 3). The results of our inquiry show that the time between onset and identification of the three subtypes varies, which might lead to errors in our analysis results. For analysis, we propose that data from cohort samples be examined.

Despite increasing full vaccination coverage of contacts over time (Delta: 44.33%, BA.1: 79.77%, BA.2: 68.72%), vaccinations are ineffective at preventing VOC infection. We discovered that fully vaccinated people had a lower risk of

infection than people not fully vaccinated (OR: 0.32 (95%CI: 0.21-0.48)), which was similar to the results reported by several studies that suggested Sinopharm or Sinovac reduced infection by 50% to 70%[38, 39]. However, we had not observed vaccine efficacy for controlling Omicron infection, including BA.1 and BA.2 subtypes, despite a rate of fully vaccinated individuals exceeding 65% and a rate of booster-vaccinated individuals exceeding 30%. Studies on Sinovac and Sinopharm vaccines showed that two doses of Sinovac did not produce sufficient antibodies to fight Omicron, as well as a "significant reduction" in the efficacy of a booster vaccination[40, 41]. The results displayed above matched our findings, indicating that there was very little vaccination efficacy in controlling Omicron. Another possibility is that this study examined different vaccine manufacturers and only included approximately 30 instances with Omicron BA.1 and BA.2. Because there were only about 30 instances of Omicron BA.1 and BA.2 infection, certain limitations are assumed. As a result, a larger sample is needed for further investigation of the vaccine's real-world ability to suppress Omicron.

In this study, the transmission characteristics and vaccination effects of Delta and Omicron variants in China were examined for the first time. Omicron outbreaks are more difficult in terms of transmission prevention and infection control than Delta outbreaks, owing to Omicron's higher communicability, shorter transmission generation time, and lower vaccination efficiency. All the aforementioned factors present additional obstacles for the dynamic zero-COVID policy. To reduce excess mortality owing to medical resource constraints in future pandemics, it may be

necessary to increase booster vaccination rates and make treatments more widely available in circumstances where vaccination failed to stop transmission.

Methods

Ethics

This study was approved by the institutional ethics committee of the Zhuhai Center for Disease Control and Prevention (CDC), Guangdong, China. Written consent was obtained from patients or their guardian(s) when samples were collected. The Hunan Provincial CDC, China and the Xiamen CDC, Fujian, China have a data-sharing agreement of infections and contacts in 2021-10 Delta outbreaks and 2022-2 Omicron BA.2 outbreaks.

Data collection

Based on data at the individual resident level, we compared three localized outbreaks caused by different variants of SARS-Cov-2, which were confirmed by National Center for Disease Control and Prevention, China. The analysis of the Omicron BA.2 outbreak was based on data reported by Xiamen CDC (from March 10 to April 6, 2022). Omicron BA.1 outbreak analysis was based on data reported by Zhuhai CDC (from January 8 to January 21, 2022). Delta outbreak analysis was based on data reported by Hunan provincial CDC (from July 22 to August 15, 2021)[15]. Three outbreaks with a different variant in each were all identified as related to other COVID waves in various Chinese cities during the same period and introduced by travelers.

Infections included in this study were identified through the national surveillance system of China, which records all individual-level information of each infection, including onset date, possible exposure date, reverse transcription-polymerase chain reaction (RT-PCR) date, and RT-PCR results. All confirmed infections were required have had at least two positive SARS-CoV-2 RT-PCR, with a detection reagent produced by two different manufacturers, and reconfirmed by laboratory at the parent institution, regardless of symptoms. In addition, detected N gene or ORF gene with Cycle threshold (Ct) value below 40 was recorded as positive from specimens from different locations, including nasopharyngeal, pharyngeal, and anal swabs. The categorization of infected people into five groups by symptom severity was consistent with that of the National Health Commission of China[16]: asymptomatic infections (not showed any symptoms within 14 days), mild cases (showed mild clinical symptoms and no pneumonia manifestation on imaging), moderate cases (showed pneumonia manifestation in imaging, fever, and respiratory tract symptoms), and severe cases, (severe symptoms and an acute condition[16]). Cases fitting in the last category were only diagnosed in Delta outbreak.

Contacts were defined as individuals who engaged, without effective protection, with infected patients at least 2 days before the first onset of symptoms, or, in an asymptomatic infection, 2 days before positive infective specimen sampling, as per COVID guidelines[16]. Contact tracing was based on infection recall, social relationships, and technological footprints, including localization of mobile communication base stations and payment logs. Contact tracing data was collected by

local health workers who received systematic training before this outbreak, containing contact vaccination status, contact frequency, contact scenario (including classroom, house, bus), and the relationship between infected individuals and their contacts.

Vaccination status

In this study, if vaccination dose were provided at the time of the epidemiological investigation, infections and contacts were fully vaccinated with a booster dose against SARS-Cov-2 based on the local health code system. The system recorded the number, date, and manufacturers of the vaccines administered to the vaccinated population, linked with each clinic and hospital that provided marketed vaccines in mainland, China. Vaccination status of infected individuals and contacts accounted for the same 14-day lag: not fully vaccinated (vaccinated with only one dose), fully vaccinated (14 days after the second dose), and booster dosed (14 days after the booster dose); however, only inactivated vaccine (Sinovac, Sinopharm, Kangtai and their branch) data were consistent. Individuals who took one dose of CanSinoBIO and or three doses of RBD-Dimer were considered to be fully vaccinated after 14 days. Furthermore, mixed dosing was defined as having taken vaccines from different manufacturers; no distinction was made regarding the specific branch of vaccine manufacturer. Breakthrough infection was defined based on vaccine status and infection category. Asymptomatic breakthrough infection was defined as SARS-Cov-2 infection without symptoms in those who have completed two doses of vaccine after 14 days. Symptomatic breakthrough infection was defined as SARS-Cov-2 infection with symptoms 14 days after second dose of vaccine[3]. In addition, the dates of public

265 emergency use authorization for vaccines varied among the age groups, divided into
266 individuals over 18 years (July 2020), 12-17 years old (July 2021), and 3-11 years old
267 (October 2021)[17, 18]. And booster dose only authorized in over 18 years people[19].

268 **Characterization of outbreaks**

269 We extracted the date of onset for symptomatic infections from SARS-Cov-2
270 infection data to estimate serial interval (SI), which is the time difference between
271 symptom onset between the infecting and infected individuals. Incubation period (IP)
272 was defined as the time difference between exposure and onset for symptomatic
273 infections; we selected possible exposure date ranges and dates of onset to estimate IP
274 through the Epitrix package[20] of R (R Foundation, Vienna, Austria), because date of
275 exposure cannot be clearly identified and most infections involved multiple
276 exposures. The estimation of generation time (GT) was limited by this issue as well,
277 so we simply enumerated all possible infecting-infected pairs and estimated GT
278 probability. We also calculated transmission generation (TG), defined as the
279 difference between positive infection test dates of the infecting and infected
280 individuals, including both symptomatic asymptomatic infections in the Zhuhai
281 outbreak. GT, TG, and SI were fitted for gamma distribution by the MASS package[21]
282 of R. We also calculated the difference between the IP and SI at the individual
283 resident level. To compare the transmissibility of varied variants of SARS-Cov-2, we
284 estimated effective reproductive number (R_{eff}) of each outbreak by maximum
285 likelihood at the increased stage[22], from first infection onset to peak of outbreak, with
286 minor public health and social measures (PHSMs) implemented.

Statistical analysis

Vaccine effectiveness on Ct value was estimated with the use of two-tailed Kruskal-Wallis test, comparing the difference of Ct value between incompletely vaccinated, fully vaccinated, and booster dosed individuals. Similar analyses were performed with variants of SARS-Cov-2. Using nested ANOVA, we assessed the difference of Ct value in the context of inner vaccine status in variants that were separated on previous analysis. Logistic regression was used to estimate vaccine effectiveness against SARS-Cov-2 infection for fully vaccinated or booster dosed individuals as compared with incompletely vaccinated individuals. Odds ratios were translated to vaccine effectiveness using the following equation: vaccine effectiveness=100×(1−odds ratio). We also adjusted for age group (0-18 years, 19-64 years, and over 65 years) in conditional logistic regression to evaluate vaccine effectiveness. In regression modeling, the synthetic minority over-sampling technique was used for re-sampling in contacts and infections[23, 24], taking the unbalanced quantities of samples per outbreak into account. In this report all data analysis, visualization and modeling were performed by R version 4.1.2.

Data availability

The basic data of infections and contacts in this study are accessible at GitHub repository (https://github.com/xmusphlkg/zhuhai_omicron).

Code availability

All source code of analysis procedure is accessible at GitHub repository (https://github.com/xmusphlkg/zhuhai_omicron).

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311 Competing interests

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313 Acknowledgements

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315 Author contributions

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317 References

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323 Supplementary material